Learn to build a Generative Model from Autoencoders in PyTorch

Overview

Many of us are aware of the Discriminative models, which generally focus on predicting the data labels. In contrast, Generative Models help to generate the output, which is very similar to the input data. For example, we have some data related to cars that can be used to classify the cars, but Generative models can learn the patterns in the data and generate car features completely different than the input data.
In this project, we start by introducing Generative Models. The PyTorch framework is used to build Autoencoders on the MNIST dataset. Finally, we learn how to use Autoencoders as Generative Models followed by generating new images of digits by using the Generative Model.

Aim

To build Generative Models by using Autoencoders in PyTorch

Tech Stack

- ➔ Language: Python
- ➔ Libraries: torch, torchvision, torchinfo, numpy, matplotlib

Approach

1. Introduction to Generative Models
2. Introduction to Autoencoders
3. Buiding Autoencoders on PyTorch
4. Model training on Google Colab
5. Building Generative model by using Autoencoders

Modular code overview

```
data
  |_data_utils.py

model
  |_autoencoder_decoder_model.py
  |_auto_encoder_colab.ipynb

run.py

train.py

requirements.txt
```

Once you unzip the modular_code.zip file, you can find the following folders within it.

1. data
2. model
3. run.py
4. train.py
5. requirements.txt

1. The data folder contains data_utils.py file which is used to download and transform the data. It will download and store the data in respective folder

2. The model folder contains autoencoder_decoder_model.py and the colab notebook

3. The train.py files contains the code for model training. The run.py is main file in which all functions are called

4. The requirements.txt file has all the required libraries with respective versions. Kindly install the file by using the command **pip install -r requirements.txt**
   Note: please use cuda versions of torch if CUDA is available

Project Takeaways

1. What are Discriminative Models?
2. What is the Generative Model?
3. Use of Generative Models

4. Introduction to Autoencoders
5. Understanding architecture of Autoencoders
6. What is Encoder?
7. What is Decoder?
8. What is the need for the Decoder layer?
9. Loss function in Autoencoders
10. How to download and transform data in PyTorch?
11. How to calculate input image size for each layer?
12. How to build Encoder Layer from scratch in Pytorch?
13. How to train the model on Colab with GPU?
14. How Autoencoders are used as a Generative Model?
15. How to generate new data by using the Generative Model?